

Karnofsky Performance Status (KPS) at presentation, tumor location, and radiation therapy. There was a trend for patients > 3 years old ($n = 31$) to have higher survival rates than those ≤ 3 years ($p = 0.084$). Patients with a gross total resection (GTR) tended to have better survival rates than those with a subtotal resection (STR), though this was not a statistically significant finding ($p = 0.30$). The median survival was 168 months for those with a GTR (range: 8 to 269) as compared to 36 months for those with STR (range: 0 to 363). The use of chemotherapy had no impact on survival ($p = 0.51$).

Patients with low-grade (1 and 2 of 4) lesions had a 5-year survival rate of 58% as compared to 34% for those with high-grade (3 and 4 of 4) lesions ($p < 0.01$). Patients with supratentorial tumors had a 5-year survival rate of 77% as compared to 41% for those with infratentorial tumors ($p < 0.02$). A KPS ≥ 70 at presentation was associated with better overall survival ($p < 0.01$).

The use of radiotherapy was also associated with a significant improvement in disease specific and overall survival on multivariate analysis ($p < 0.001$). The disease specific survival rate at 5 years was 41% with radiation versus 7.1% without. The five-year overall survival rate with radiation was 60% as compared to 35% without. Time from surgery to radiation (< 6 weeks or ≥ 6 weeks) had no impact on survival ($p = 0.77$). Patients whose radiation was delayed received a median dose of 55.8 Gy (range 50.4 to 59.4 Gy, $n = 11$) while patients whose radiation was not delayed received a median dose of 50.4 Gy (range 43.2 to 59.4 Gy, $n = 29$).

Conclusions: Radiation therapy, low-grade histology, supratentorial location and high KPS were all associated with improved overall survival in this analysis. A trend for better survival was found in patients having a GTR versus STR, however this did not reach statistical significance. This may be due to the relatively small numbers of patients having adequate postoperative imaging to accurately assess extent of resection. Timing of postoperative radiotherapy was not found to significantly impact outcome in this group of patients.

1090 Improved Outcomes After Stereotactic Radiotherapy for Craniopharyngioma

M. Koshy, C. G. Morris, N. P. Mendenhall

Radiation Oncology, University of Florida, Gainesville, FL

Purpose/Objective: The role and timing of radiotherapy (RT) in the management of craniopharyngioma is controversial. The purpose of our study was to assess the overall survival, local recurrence, and toxicity rates after RT.

Materials/Methods: Thirty-three patients were treated between 1983 and 2001 and followed from 1 month to 19.3 years (median, 4.8 years). There were 10 pediatric patients and 23 adult patients. Conventional RT was used in 13 patients, and stereotactic radiotherapy (SRT) in 15 patients. The dose ranged from 49 to 60 Gy (median, 54 Gy). Twelve patients received RT after initial surgery as part of the initial therapeutic plan, and 16 patients were irradiated for recurrent tumor after multiple surgical interventions. Five patients underwent 32P radionuclide instillation for salvage therapy.

Results: The local control and actuarial survival rates at 5 years after RT were 65% and 65%. At last follow-up, 16 patients (65%) were alive and disease-free, 9 died with disease, and 3 died of intercurrent disease. Median time to recurrence was 18 months (range 1 month to 18 years). There were 6 cause-specific deaths associated after RT for recurrent disease and 3 cause specific deaths in those who received RT after initial surgery. Seven patients had a recurrence after conventional RT, and 4 after SRT. Two-thirds of the patients had diabetes insipidus and panhypopituitarism after surgery, 3 required VP shunts, and 2 suffered from seizures. Long-term treatment complications included emotional lability and cognitive deficits after conventional RT. No similar complications were observed in patients who received SRT. Five patients received 32P radionuclide instillation after undergoing multiple surgical interventions for recurrent tumor; three had received prior RT. Three of 5 patients died with disease and two were lost to follow-up.

Conclusions: SRT appears to be as effective as conventional RT in obtaining tumor control. Furthermore, neuropsychologic complications appear to be less frequent after SRT than with conventional RT. Because of the high likelihood of progression after subtotal resection and the increased likelihood of surgical complications with each operation, we recommend SRT after initial subtotal resection for craniopharyngiomas.

1091 Uveal Melanoma Recurrent After Fractionated Proton Beam Therapy: Comparison of Survival in Patients Treated with Re-irradiation or With Enucleation

L. Marucci,¹ M. Ancukiewicz,² A. M. Lane,³ J. M. Collier,² E. S. Gragoudas,³ J. E. Munzenrider²

¹Radiation Oncology, Regina Elena Research Institute, Rome, Italy, ²Radiation Oncology, Massachusetts General Hospital, Boston, MA, ³Retina Service, Massachusetts Eye and Ear Infirmary, Boston, MA

Purpose/Objective: To retrospectively compare survival in recurrent uveal melanoma, between patients treated by enucleation and by a second course of fractionated proton beam radiation therapy (PBRT).

Materials/Methods: Tumor recurrence was documented in 73 of the 2252 patients treated with PBRT for uveal melanoma between 1/1982 and 12/1998 at the Harvard Cyclotron Laboratory, in collaboration with the Radiation Oncology Department, Massachusetts General Hospital and the Retina Service, Massachusetts Eye and Ear Infirmary. Treatment for recurrent tumor was determined by tumor configuration and dimensions, and by the patient's performance status and treatment preference. Thirty-one patients received a second course of PBRT and 42 were enucleated. Mean patient age was 59 years for the entire group (range 25 – 83 years), and 56 and 61 for those enucleated and re-irradiated, respectively ($p = 0.09$). Both primary and recurrent tumors were larger in patients undergoing enucleation: 0.73 ± 0.60 vs. 0.38 ± 0.37 cc ($p = 0.008$) at the time of initial PBRT and 0.68 ± 0.71 vs. 0.33 ± 0.30 cc ($p = 0.025$) at second irradiation or enucleation (means $\pm$ S.D.). Tumor location and presence or absence of ciliary body involvement did not differ significantly between groups. All re-irradiated patients had 5 treatments initially: 28 received 70 CGE and 3 received 50 CGE. For the second course, 30 patients had 70 CGE in five fractions, and one had 48 CGE in 4 fractions. One patient received a third course of 70 CGE in five fractions. Median interval between the initial and the second course of PBRT was 25 months, and between initial PBRT and enucleation was 36 months ($p = 0.24$). Median follow up after enucleation or re-treatment was, respectively, 79 and 59 months ($p = 0.52$).

Results: At last follow-up, 17 patients in the re-irradiated group survived without metastasis, 1 was alive with metastasis, 10 were dead of metastasis and 3 of other causes. In the enucleated group, 15 patients were alive without disease, one was alive with metastasis, 25 were dead of metastasis and 2 of other causes. Median survival duration in enucleated and re-irradiated groups was, respectively, 42 and 90 months ($p = 0.040$, log-rank test). Median time free of metastases was 38 months in enucleated and 97 months in re-irradiated patients ($p = 0.028$, log rank-test). At 5 years after enucleation or re-irradiation, the probability of overall survival was, respectively, 36% and 63%; the probability of freedom from metastases was, respectively, 31% and 66%. However, the differences in risk of all cause mortality and of onset of metastasis were not significant after adjustment for recurrent tumor volume in the proportional hazards model. Nine eyes had been enucleated in the irradiated group, 5 for local recurrence and 4 because of intractable pain. The probability of survival with eye retention at 5 years was 40% for the re-irradiated group.

Conclusions: This retrospective analysis suggests that survival in re-irradiated patients is not compromised by administration of a second course of PBRT for recurrent uveal melanoma. The study also shows that patients with uveal melanoma recurrent after high dose fractionated proton beam therapy can be re-irradiated with the same modality using the same dose as given initially, with acceptable morbidity, and with a reasonable expectation of eye retention.

1092 Effectiveness of Fractionated Stereotactic Radiation Therapy for Uveal Melanoma

K. Muller,¹ G. Luyten,² P. Nowak,¹ J. Marijnissen,¹ C. de Pan,¹ P. Levendag¹

¹Radiotherapy, Erasmus MC, University Medical Center, Rotterdam, Netherlands, ²Ophthalmology, Erasmus MC, University Medical Center, Rotterdam, Netherlands

Purpose/Objective: Uveal melanoma has traditionally been treated by enucleation. Since 1970, various eye sparing treatment modalities have been developed with as aim an optimal sparing of visual function. Brachytherapy appears to be a good alternative for enucleation in the treatment of small uveal melanoma, according to recent comparative studies. For medium sized tumors, proton beam and helium ion radiation techniques are widely used. Radiosurgery for uveal melanomas has been introduced in 1987 with excellent results of local control. The most important disadvantage of single fraction high dose radiosurgery is an enhanced complication probability, which causes the loss of radiobiological sparing if critical structures are adjacent to the tumor. Fractionated stereotactic radiotherapy reduces the effective dose in the surrounding tissues with a delivery of an equivalent effective dose to the tumor.

This study analyses the effectiveness and side effects of fractionated Stereotactic Radiation Therapy (f.SRT) for uveal melanoma in a prospective, nonrandomized, self-controlled clinical trial.

Materials/Methods: Between 1999 and 2003, 38 uveal melanoma patients (21 male, 17 female) with mean tumor thickness of 6.4 mm were included in the study. Informed consent was obtained from all patients. The mean follow-up was twenty-five months, with a minimal follow-up of 10 months. No patients were lost to follow-up. A total dose of 50 Gy, prescribed to the 80% isodose line, was given in 5 daily fractions of 10 Gy by LINAC-based SRT. A blinking light in front of the unaffected eye and a camera to monitor the position of the diseased eye are fixed to a non-invasive relocatable stereotactic frame. The main primary outcome measures were local tumor control and visual acuity at 3, 6, 12 and 24 months respectively and acute toxicity (peri- and intraocular side-effects).

Results: After three months (38 patients) the local control was 100%, after 12 months (32 patients) one patient was treated by transpupillary thermotherapy because the tumor size did not decrease and after 24 months (15 patients) no recurrences were seen. The visual acuity gradually declines from a mean of 0.21 at diagnosis to a mean of 0.06 two years after therapy. The acute side effects after three months were: conjunctival symptoms (10), loss of eye lashes (6), visual symptoms (5), fatigue (5), dry eye (1), cataract (1), and pain (4). One eye was enucleated 2 months post f.SRT.

Conclusions: Preliminary results demonstrate that fractionated SRT is an effective and safe treatment modality for uveal melanoma with excellent local control and mild acute side effects. The follow-up period is too short to discuss meaningfully the risk for long term side effects.

1093 Decreased Incidence of Neovascular Glaucoma by Sparing Anterior Structures of the Eye for Proton Beam Therapy of Ocular Melanoma

R. R. Rajendran,¹ V. Weinberg,¹ I. K. Daftari,¹ J. R. Castro,¹ J. M. Quivey,¹ D. H. Char,^{2,3} T. L. Phillips¹

¹Department of Radiation Oncology, University of California San Francisco, San Francisco, CA, ²The Tumori Foundation, San Francisco, CA, ³Department of Ophthalmology, Stanford University, Stanford, CA

Purpose/Objective: We previously hypothesized that sparing the anterior segment of the eye would reduce the incidence of neovascular glaucoma (NVG) (Daftari, I. K., et al., IJROBP 1997, 39:997-1010; Meecham, W.J., et al., Arch Ophthalmol 1994, 112:197-203). To test this hypothesis, we examined the incidence and contributing factors of NVG in patients treated with proton beam therapy at the Crocker Nuclear Laboratory faculty in Davis, California using techniques for sparing anterior structures of the eye.

Materials/Methods: Using anterior structure sparing techniques, 346 patients were treated with proton beam therapy between June 1996 and November 2003 and included in the ocular melanoma database of the Tumori Foundation. Twenty-four patients were excluded due to prior surgical resection and 3 patients because they were treated for recurrences after ¹²⁵I plaque therapy. This resulted in 319 evaluated patients treated with anterior structure sparing techniques. The patients in the study received 56 GyE in 4 fractions. The primary endpoint of this study is the occurrence of NVG. The median follow-up is 24.7 months. The log rank test was used to compare distributions of the occurrence of NVG. The likelihood ratio test (LLRT) from Cox's proportional hazard model was used to compare models to identify independent predictors of outcome.

Results: NVG occurred among 39 of the patients. The probability of remaining free of NVG at 3 years was 89%. The smaller the tumor diameter was, the better the outcome with 3 year estimates of NVG for the quartiles <7.5 mm, >7.5 mm - 10.5 mm,